Yue Xue (薛岳)

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Google Scholar: https://scholar.google.com/citations?user=6cdLQQQAAAAJ

SUMMARY

- Senior Security Engineer focused on blockchain security, software security, and LLM applications in security.
- An early LLM4Sec researcher and builder: co-author of foundational, highly-cited works (GPTScan, PropertyGPT), advancing a security methodology combining LLMs with program analysis/formal verification, and building Finite Monkey to industrialize AI smart contract auditing at scale.
- Owned internal anti-money laundering (AML) initiatives at OKX with end-to-end on-chain compliance experience, from fund monitoring and risk identification to alerting/interdiction and closed-loop handling.
- Experienced in smart contract auditing, static analysis, formal verification, and on/off-chain monitoring systems.
- Awards: ETHSecurity Badge holder; NDSS 2025 Distinguished Paper Award; ACM SIGSOFT Distinguished Paper Award (FSE 2024); Finite Monkey ranked #2 on Secure3, earning USD 68,570 in cumulative bounties.

EDUCATION

Guangzhou University (GZHU) 2020.09 – 2023.06

M.S. in Computer Technology (Academician Fang Binxing Class)

· Research: blockchain security, smart contract security, LLM for security

University of Electronic Science and Technology of China (UESTC) 2011.09 – 2015.06

B.E. in Electrical Engineering and Automation

· Focus: electrical engineering, automatic control, and core engineering coursework

EXPERIENCE

CertiK 2025.11 – Present

Head of AI Research

· **Product engineering:** Lead developer of AI Auditor, a multi-language AI vulnerability scanning SaaS product spanning code understanding, threat modeling, vulnerability discovery, validation, and reporting.

· **Core capabilities:** Build finding deduplication, validation, and self-evolving workflows; optimize model usage, context reuse, and parallel execution to reduce costs and improve reliability.

· **Evaluation:** Built evaluation and regression workflows. Against 644 medium-or-higher findings in historical ACC human audit reports, achieved 80.1% exact-match coverage and 88.2% coverage including partial matches (August 27, 2026).

· **Competition results:** In an October 2026 Sherlock contest, the product tied for #1 among AI entrants in valid findings (22), tied for #1 overall in unique findings (7), and ranked #1 overall in coverage efficiency.

OKX 2022.04 – 2022.10, 2024.05 – 2025.11

Senior Security Engineer / AML Risk Operations Lead

· **AML risk control:** Owned ATMS (Asset Transaction Monitoring System) development and operations for first-line external-deposit risk control, covering illicit-fund detection and interception across CeFi deposits and wallet transactions.

· **Contract asset security:** Managed all internal OKX contracts through a unified platform and led the transfer of contract control to the asset management team. Established multisig initiation and asset management coordination for critical calls, safeguarding contract funds with zero migration incidents.

· **On-chain investigations:** Led high-risk address tracing, linked-address expansion, intelligence maintenance, and risk assessment; coordinated manual investigations, fund-flow analysis, and risk reporting.

· **Team leadership:** Led a four-person team building smart contract monitoring for state and transaction anomalies; contributed to contract audits and security tooling.

MetaTrust 2022.10 – 2025.08

Security Supervisor / Consultant

- **Audit delivery:** Led delivery and team coordination for 50+ completed smart contract audit engagements, including repeat audit rounds, improving audit workflows and quality control.
- **Detection tools:** Developed 60+ Slither detectors and optimized nearly 100 rules, reducing the false-positive rate below 25%.
- **AI security research:** Contributed to GPTScan and independently built PropertyGPT, combining LLMs with program analysis and formal verification.
- **Research collaboration:** Collaborated with NTU research teams and co-authored publications.

C# Developer 2016.12 – 2019.01

- Built full-stack web features and maintained databases
- Developed and maintained enterprise systems

PROJECTS & TOOLS

LLM4Sec Audit Tools Ongoing

End-to-end audit toolchain

- Code Audit Hinter (IDE security plugin, 39 stars, 2025.03–Present): RAG-powered audit hints for VSCode and Cursor
- Finite Monkey Engine (360 stars, 2024.04–Present): AI vulnerability scanning product built on a task-driven LLM audit engine
- TxAnalyzer (GitHub, 2026.05–Present): agent skill and CLI for blockchain attack transaction root-cause analysis, with EVM/Solana artifact collection
- AI Audit Result Evaluator (private, 2025.05–Present): automated judging for contest findings vs. ground truth
- Code Semantic Search (private, 2024.12–Present): LLM-based codebase understanding and retrieval

Falcon 2022.12 – 2024.05

Enhanced Solidity static analysis framework

- Extended Slither with dozens of new detectors and improved precision
- Supports DeFi price manipulation, centralization risk, and complex logic bug detection

Secure3 Audit Practice and Product Results Ongoing

Manual and AI-assisted auditing

- Completed approximately 20–30 Secure3 audit engagements, using manual auditing and AI-assisted analysis to identify and validate vulnerabilities.
- Built Finite Monkey, which ranked #2 on Secure3, earning USD 68,570 in cumulative bounties and demonstrating AI scanning performance in real-world audits.

PUBLICATIONS

1. Y. Sun, D. Wu, **Yue Xue**, H. Liu, H. Wang, Z. Xu, X. Xie, Y. Liu. "GPTScan: Detecting Logic Vulnerabilities in Smart Contracts by Combining GPT with Program Analysis." Proc. ICSE, 2024.
2. Y. Sun, D. Wu, **Yue Xue**, H. Liu, W. Ma, L. Zhang, Y. Liu, Y. Li. "LLM4Vuln: A Unified Evaluation Framework for Decoupling and Enhancing LLMs' Vulnerability Reasoning." arXiv preprint arXiv:2401.16185, 2024.
3. W. Ma, D. Wu, Y. Sun, T. Wang, S. Liu, J. Zhang, **Yue Xue**, Y. Liu. "Combining Fine-Tuning and LLM-based Agents for Intuitive Smart Contract Auditing with Justifications." Proc. ICSE, 2025.
4. Y. Liu, **Yue Xue***, D. Wu, Y. Sun, Y. Li, M. Shi, Y. Liu. "PropertyGPT: LLM-driven Formal Verification of Smart Contracts through Retrieval-Augmented Property Generation." NDSS, 2025.
5. K. Li, **Yue Xue**, S. Chen, H. Liu, K. Sun, M. Hu, H. Wang, Y. Liu, Y. Chen. "Static Application Security Testing (SAST) Tools for Smart Contracts: How Far Are We?" Proc. ACM Softw. Eng. (FSE), 1, 1447–1470, 2024.
6. **Yue Xue**, J. Fu, S. Su, Z. A. Bhuiyan, J. Qiu, H. Lu, N. Hu, Z. Tian. "Preventing Price Manipulation Attack by Front-Running." Proc. AAIS, 309–322, 2022.
7. M. Xie, M. Hu, Z. Kong, C. Zhang, Y. Feng, H. Wang, **Yue Xue**, H. Zhang, Y. Liu, et al. "DeFort: Automatic Detection and Analysis of Price Manipulation Attacks in DeFi Applications." Proc. ISSTA, 2024.
8. **Yue Xue**, D. Fan, S. Su, J. Fu, N. Hu, W. Liu, Z. Tian. "A Review on the Security of the Ethereum-Based DeFi Ecosystem." CMES–Computer Modeling in Engineering & Sciences, 139(1), 2024.
9. S. Su, Y. Tan, **Yue Xue**, C. Wang, H. Lu, Z. Tian, C. Shan, X. Du. "Detecting Smart Contract Project Anomalies in Metaverse." Proc. IEEE MetaCom, 2023.

10. Z. Kong, C. Zhang, M. Xie, M. Hu, **Yue Xue**, Y. Liu, H. Wang, Y. Liu. "Smart Contract Fuzzing Towards Profitable Vulnerabilities." Proc. ACM Softw. Eng. (FSE), 2, 153–175, 2025.

11. S. Su, **Yue Xue**, L. Lin, C. Wang, H. Lu, J. Qiu, Y. Sun, Y. Liu, Z. Tian. "Smart Contract Firewall: Protecting the On-Chain Smart Contract Projects." Proc. IEEE GLOBECOM, 5122–5128, 2023.

12. Y. Sun, S. Ding, L. Tan, **Yue Xue**, B. Li, C. Huang, S. Xiong. "Beyond Stars and Commits: A Fair and Manipulation-Resistant Framework for Measuring Developer Contributions." Proc. ACM FSE Companion, 2025.

13. J. Gao, **Yue Xue**, X. Xie, J. Cao, S. Thant, E. Lee, B. Xu. "Understanding Codebase like a Professional! Human−AI Collaboration for Code Comprehension."ICPC '26.

14. **Yue Xue**. "When and How Context Rot Appears in Coding Agents: A White-Box Study of Agent Skills in Code Auditing." arXiv '26 (preprint).

15. **Yue Xue**. "Chiral Analysis of Smart Contracts: Detecting Vulnerabilities from Relational Inconsistencies Across Business Paths."arXiv '26 (preprint).

16. Z. Kong, W. Xia, C. Wang, **Yue Xue**, Y. Lu, P. Li, S. Li, Z. Cao, Y. Liu. "Knowdit: Agentic Smart Contract Vulnerability Detection with Auditing Knowledge Summarization."arXiv '26 (preprint).

17. R. Liang, J. Chen, X. Li, H. Gu, Y. Feng, **Yue Xue**, C. Wu, Y. Liu. "EvoPoC: Automated Exploit Synthesis for DeFi Smart Contracts via Hierarchical Knowledge Graphs."arXiv '26 (preprint).

* Co-first author.

ACADEMIC SERVICE

- Reviewer: IEEE Transactions on Software Engineering (TSE); ACM Transactions on Software Engineering and Methodology (TOSEM); ACM Transactions on Privacy and Security (TOPS)
- Reviewer: 1st Workshop on Interpreting Agent Behavior (IAB), NeurIPS 2026.

INDUSTRY WRITING

- Write on AI auditing, agent methodology, and on-chain attacks. The methodology series has been republished by DeFiHackLabs, translated by LearnBlockchain, and included in a yBorg preparation reading list.
- **AI Auditing Methodology: Everything You Need to Know (Part I)**: English on X (51,212 views); Chinese translation (737 reads).
- **AI Auditing Methodology: Everything You Need to Know (Part II)**: English on X (16,812 views); Chinese translation (566 reads).
- **AI Auditing Methodology: Agent Self-Evolution – Drift, Reverse Evolution and Solutions** (June 2026); X version (19,936 views).
- **CertiK AIAuditor: Some Thoughts on AI Audit Methodology** (August 2026).
- **Why Your Skill Keeps Ignoring Instructions** (August 2026).

View and read counts retrieved October 14, 2026.

MEDIA INTERVIEW

- Interviewed expert: International Financial News, August 18, 2026, Repeated Boundary Violations: Is AI Getting Out of Control?, discussing AI agent safety boundaries and safeguards.

AWARDS

- NDSS 2025 Distinguished Paper Award: PropertyGPT
- ACM SIGSOFT Distinguished Paper Award (FSE 2024): SAST Tools for Smart Contracts
- GLOBECOM 2023 Best Paper Award: Smart Contract Firewall
- ETHSecurity Badge holder
- 5th place, SCAN 2026 Digital Asset Tracing Challenge; member of CertiScan, representing CertiK
- Finite Monkey ranked #2 on Secure3, earning USD 68,570 in cumulative bounties